Number Continuum Based on Line Continuity Axiom

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Some mathematicians insisted that we should divorce the concept of the real numbers from its geometric origins, so the real numbers, especially irrational numbers , should not be constructed on geometry,

Weierstrass’ s reason[1]:

Inasmuch as it was apparent to Weierstrass that (geometric)intuition could not be trusted, he sought to make the bases of his analysis as rigorously and precisely formal as possible...In order to secure logical exactitude, Weierstrass wished to establish the calculus (and the theory of functions) upon the concept of number alone, thus separating it completely from geometry.

Dedekind’ s reason [2] :

Just as negative and fractional rational numbers are formed by a new creation, and as the laws of operating with these numbers must and can be reduced to the laws of operating with positive integers, so we must endeavor completely to define irrational numbers by means of the rational numbers alone.

However, their reasons aren’ t proofs on showing why it is illegitimate or impossible to constructed irrational numbers on geometry, and after

[1] Carl B. Boyer, The History of the Calculus and its Conceptual Development, 285

[2] Richard Dedekind, Essays on the Theory of Numbers, P5

read the following[3], I decided to try instead.

> Sometimes authors and lecturers on analysis insist that students must not use geometrical intuition in developing the fundamental concepts of analysis or in constructing proofs in analysis. Such a prohibition appears to be consonant with Felix Klein's description in 1895 of the developments due to Weierstrass, Cantor and Dedekind, namely the *Arithmetisation of Analysis*.
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> However, such advice is impossible to implement and is in any case untrue to the origins of the subject. We can hardly conceive of a Dedekind cut, for example, without imagining a 'real line', and such imagining was certainly part of Dedekind's own thought. We have eyes, and we have imaginations with which to visualise, and such visualisation is central to much of the development of analysis. Every development of the real number system is a way of formalising our intuitions of the points on an endless straight line. We cannot conceive how the theory of real functions could have developed had there been no graphs drawn.

I admit that geometrical intuition is sometimes misleading, so my construction should be aware of not rely on it but rather on correct geometric conclusions or geometric axioms.

Dedekind found the essence of the continuity of a line in the nature of the division of the line by a point. He saw that in any division of the points of a line into two classes such that every point of the one is to the left of every point of the other, there is one and only one point which produces this division. This is not true of the ordered system of rational numbers. This, then, was why the points of a line formed a continuum, but the rational numbers did not. As Dedekind expressed it, "By this commonplace remark the secret of continuity is to be revealed",

"The assumption of this property of the line is nothing else than an axiom by which we attribute to the line its continuity, by which we find

[3] R. P. Burn, Numbers and Functions: Steps into Analysis, P330

continuity in the line" .

Logically and naturally, I think Dedekind should immediately put forward the property that could characterize the number continuum after inspired by the line continuity axiom rather dive into the cut of the rational numbers like that written in his pamphlet Continuity And Irrational Numbers. That is, he should put forward the ***number continuum axiom***[4] or the completion axiom immediately after inspired by the line continuity axiom:

The domain R possesses also continuity; i. e., the following theorem is true: If the system R of all numbers breaks up into two classes A_1 , A_2 *such that every number* α_1 *of the class* A_1 *is less than every number* α_2 *of the class* A_2 *then there exists one and only one number* α *by which this separation is produced.*(slightly modified from the original)

The existence of the separation produced by no rational number disproves the rational numbers to be the continuum , so the question becomes in what way could the domain of rational numbers be rendered complete to form a continuous domain?

Every rational number could be represented by a unique point on a straight line, but there are points on the line which don' t correspond rational numbers, if we call these points irrational points , and assume there is one and only one non-rational number, name it the irrational

[4] Richard Dedekind, Essays on the Theory of Numbers, P9

number, corresponding to every irrational point, and state that the irrational number is greater(or less) than the numbers for which the points they correspond to lies to the left(or right) of the irrational point. If we call both rational numbers and irrational numbers real numbers , then by this definition we establish a perfect correspondence between points and real numbers, any two distinct points is either to the left or right to each other on a horizontal line, likewise, any two distinct real numbers is either less or greater than each other , that is, there is an isomorphism between the set of points on a number line and the set of real numbers, since the line preserve the line continuity axiom, the set of real numbers should preserve the number continuum axiom, in other words, the set of real numbers is the number continuum.

The number continuum axiom ensures the least upper bound property of real numbers, I prove it as following :

If S is a non-empty subset of the number continuum R and is bounded above in R, we collect all those real numbers greater than every of S in C and all real numbers not in C to set B, according to the number continuum axiom there is one and only one number d in R producing B and C. Since every number of B is not greater than every of S, so the upper bound of S is also B, and since every number of A is also in B, so the upper bound of B is also S, to sum up, B and S share the same least upper bound, so a non-empty subset of the number continuum which is

bounded above has the least upper bound in the number continuum. The least upper bound property ensures Archimedean property of real numbers.

Theorem 1.2.8 (The Archimedean Property for the Real Numbers). *If α and β are positive real numbers, then there is some positive integer n such that $n\alpha > \beta$.*

In the proofs in this section we will assume that real numbers obey the rules of arithmetic for inequalities.

PROOF (By contradiction.) Suppose that $n\alpha > \beta$ did not hold for any integer n, that is, for all integers n we have $n\alpha \leq \beta$. Then the set $A = \{n\alpha \mid n \in N\}$ is bounded above by β. By Theorem 1.2.4 there is a least upper bound for A, which we will call λ.

Now since α is positive, $\lambda - \alpha < \lambda$; therefore, $\lambda - \alpha$ is not an upper bound for the set A. Thus, for some integer m we must have $\lambda - \alpha < m\alpha$ or, equivalently, $\lambda < (m+1)\alpha$. But this contradicts the choice of λ as an upper bound for A. ■